

Counting the Smallest Polyiamond Containers of Every n-Iamond

The sequence $a(n)$ counts the distinct minimum-size connected polyiamonds that contain every free n -iamond, up to the rotations, reflections, and translations of the triangular lattice. Each figure shows ONE representative minimal container for that n ; the caption gives $a(n)$ (the number of such containers) and the detail gives their common size $k(n)$ and bounding box. Container cells are filled teal. The minimum size is proved within a $\max(4, \lfloor (n+2)/3 \rfloor)$ -row by n -column search window.

$$a(1) = 1, a(2) = 1, a(3) = 1, a(4) = 1, a(5) = 4, a(6) = 5, a(7) = 39, a(8) = 13, a(9) = 26$$

Computed by Peter Exley, Jun 2026

$$a(1) = 1 \text{ [PROVED]} \quad k=1 \text{ 1x1}$$



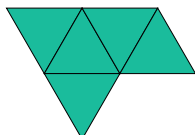
$$a(2) = 1 \text{ [PROVED]} \quad k=2 \text{ 1x2}$$



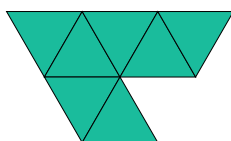
$$a(3) = 1 \text{ [PROVED]} \quad k=3 \text{ 1x3}$$



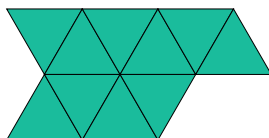
$$a(4) = 1 \text{ [PROVED]} \quad k=5 \text{ 2x4}$$



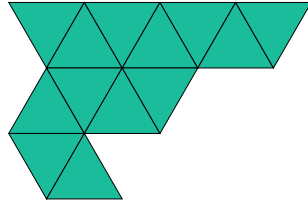
$$a(5) = 4 \text{ [PROVED]} \quad k=7 \text{ 2x5}$$



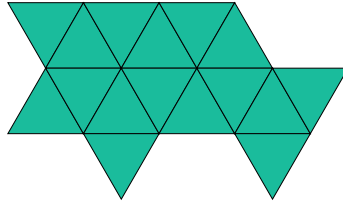
$$a(6) = 5 \text{ [PROVED]} \quad k=10 \text{ 2x6}$$



$$a(7) = 39 \text{ [PROVED]} \quad k=13 \ 3 \times 7$$



$$a(8) = 13 \text{ [PROVED]} \quad k=16 \ 3 \times 8$$



$$a(9) = 26 \text{ [PROVED]} \quad k=19 \ 3 \times 9$$

